

Spatial Representation of Large Language Models in 2D Scene

Wenya Wu

Mashang Consumer Finance Co., Ltd
Chongqing, China
sophie_wwy@pku.edu.cn

Weihong Deng

Mashang Consumer Finance Co., Ltd
Chongqing, China
weihong.deng@msxf.com

Abstract

Spatial representations are fundamental to human cognition, as understanding spatial relationships between objects is essential in daily life. Language serves as an indispensable tool for communicating spatial information, creating a close connection between spatial representations and spatial language. Large language models (LLMs), theoretically, possess spatial cognition due to their proficiency in natural language processing. This study examines the spatial representations of LLMs by employing traditional spatial tasks used in human experiments and comparing the models' performance to that of humans. The results indicate that LLMs resemble humans in selecting spatial prepositions to describe spatial relationships and exhibit a preference for vertically oriented spatial terms. However, the human tendency to better represent locations along specific axes is absent in the performance of LLMs. This finding suggests that, although spatial language is closely linked to spatial representations, the two are not entirely equivalent.

1 Introduction

The apparent proficiency of large language models (LLMs) in understanding and generating natural language suggests that they may exhibit cognitive abilities akin to those of humans, such as theory of mind and reasoning (Strachan et al., 2024; Rahimi Moghaddam and Honey, 2023; Lampinen et al., 2024; Webb et al., 2023; Gandhi et al., 2023). Consequently, the evaluation of these models has garnered increasing attention, particularly given their expanding applications across domains like code generation and translation (Hong et al., 2023), where minimizing potential errors in their responses is critical. A promising direction for the LLM industry lies in advancing embodied intelligence, which necessitates a robust capacity for spatial understanding (Fan et al., 2024; Zhang et al., 2024). While spatial reasoning is more prominent

in the multi-modal domain, where spatial phenomena are often integrated with visual information, it remains essential to investigate spatial representations grounded in natural language to further enable LLMs to support and enhance various aspects of social life.

Spatial relations, which describe the connections between physical objects, are essential for spatial understanding and play a critical role in spatial reasoning. Humans naturally use language to convey spatial relations in everyday life. Trained on extensive natural language datasets, large language models (LLMs) may encode not only spatial linguistic structures but also develop implicit representations of spatial relations, even without direct sensory inputs. Understanding the interaction between spatial language and spatial representations in LLMs can offer valuable insights into how these models process and "comprehend" spatial concepts. Recent studies suggest that LLMs have achieved acceptable proficiency in representing simple cardinal directions and planning navigation tasks (Cohn and Blackwell, 2024; Zhou et al., 2024). However, their performance remains inconsistent and is influenced by factors such as environmental complexity. LLMs tend to excel in addressing basic spatial questions but struggle with more advanced and intricate spatial concepts (Hojati and Feick, 2024). Considering that spatial representations are vital for achieving embodied intelligence and advancing toward artificial general intelligence (AGI), the sensitivity of LLMs to spatial relations in 2D space warrants more comprehensive exploration.

Building on the *CogEval* protocol recently proposed for the general evaluation of LLMs' cognitive capacities (Momennejad et al., 2023), this study aims to assess the spatial intelligence of LLMs. Specifically, we examine the structure of LLMs' representations of spatial relations between two objects within a 7*7 grid scene and evaluate the similarity of these representations to those of

humans using two spatial tasks: the spatial generation task and the spatial rating task. The central research question is whether LLMs can derive visual-like representations from textual input and coordinate descriptions in a 2D space, and to what extent their representations align with those of humans. We evaluate the spatial sensitivity of five LLMs, including state-of-the-art (SOTA) models such as GPT-4, and compare their performance to human behavior data obtained from a previous related study. The research hypothesis posits that LLMs can partially capture 2D spatial representations and exhibit certain features embedded in human spatial language.

The results reveal both similarities and differences between the spatial representations of LLMs and humans. Similar to humans, LLMs more frequently select vertically oriented spatial prepositions to describe spatial relations, as opposed to horizontally oriented terms. State-of-the-art (SOTA) models, such as GPT-4, demonstrate significant proficiency in judging spatial relations, with the exception of accurately identifying the rightward relationship. However, weaker models, such as Llama3-8B, exhibit lower spatial intelligence. Furthermore, the temperature parameter appears to have minimal impact on the models' performance, suggesting that spatial representations may be fundamental to human cognition. Nonetheless, LLMs show limitations in capturing certain subtle characteristics of human spatial cognition, such as the tendency for more precise representations along specific axes.

In summary, the main contributions of this study are as follows:

1) Adaptation of a standardized experimental paradigm: We transferred a well-established experimental paradigm from cognitive psychology, used to examine spatial representations in humans, to the evaluation of LLMs. This approach reveals the models' spatial capacities in a 2D scene, which serves as a foundational aspect of spatial intelligence required in more complex environments.

2) Comparison of spatial representations: By comparing the spatial representations of five mainstream LLMs with human behavior based on previous studies, this research provides insights into the spatial capabilities of LLMs while also contributing to an indirect understanding of human spatial cognition.

2 Related Works

2.1 Spatial representations and spatial language

Fundamental to cognition in both humans and other animals, spatial representations play a critical role in encoding the geometric properties of objects and the spatial relationships among them. These representations often encompass cognitive models or mental maps that individuals use to mentally visualize and manipulate spatial information. Spatial representations are typically derived from sensory modalities such as vision, hearing, or touch, and they provide crucial information to motor systems and language processing (Landau and Jackendoff, 1993). As a result, frequent translation occurs between spatial representations and spatial language, which generally consists of spatial words or simple phrases.

Spatial language specifically refers to linguistic expressions used to describe spatial properties such as location, orientation, direction, and distance. These expressions are integral to how individuals communicate their understanding of spatial environments. Three basic elements underpin linguistic descriptions of spatial locations: the figure object (the object being located), the reference object, and the spatial relationship between them. Spatial relationships are often encoded through prepositions such as "above" and "below," while both the figure object and the reference object are typically expressed as noun phrases denoting object names. For example, in the sentence "The apple is on the desk," "the apple" functions as the figure object, "the desk" serves as the reference object, and the preposition "on" reflects the spatial relationship between them.

In cognitive psychology, spatial language and spatial representations are intricately linked. Spatial language serves as a key mechanism through which humans convey and process information about space, while spatial representations act as mental constructs that help organize and navigate spatial relationships. It has been proposed that spatial language is grounded in the geometry of visual scenes represented in spatial cognition (Mirzaee et al., 2021). Furthermore, the articulation of spatial concepts in language may influence how they are mentally represented. Empirical evidence suggests that limited exposure to spatial language impairs individuals' performance on non-linguistic spatial tasks, with deaf children showing weaker

abilities to convey spatial relations (Gentner et al., 2013). Cross-linguistic comparisons reveal that similar spatial properties are encoded in both spatial language and spatial representations, suggesting parallels between these two systems (Munnich et al., 2001). Consequently, spatial language can be viewed as a window into the spatial representations that underlie human cognition.

2.2 Spatial understanding of LLMs

Given that LLMs are trained on vast amounts of natural language data, which inherently contains rich spatial language, it is reasonable to infer that these models may acquire a certain degree of spatial understanding. This inference aligns with the established link between spatial representations and spatial language in human cognition. Although LLMs lack access to visual or sensorimotor information, studies suggest that they can partially derive spatial representations from textual input. For instance, LLMs have shown promise in reasoning about simple cardinal directions (CDs), such as "north," "south," "east," and "west," though their performance declines with more complex CDs, such as "northeast" (Cohn and Blackwell, 2024). Additionally, LLMs demonstrate some ability to perform spatial calculations and apply spatial prepositions correctly (Bhandari et al., 2023). Prompting strategies, including Chain-of-Thought (CoT), one-shot or few-shot prompting, and advanced techniques like Visualization-of-Thought (VoT), have been shown to enhance LLMs' spatial reasoning and path-planning capabilities (Wu et al., 2024; Xu et al., 2024). Breaking complex spatial reasoning tasks into smaller, manageable subtasks also improves performance (Peng and Powers, 2024).

However, challenges remain. LLMs' representations of spatial relations can be distorted, often influenced by the hierarchical structure of the environment (Fulman et al., 2024). In many cases, models identify only the nearest cardinal directions, reflecting an associative learning mechanism rather than a robust understanding of spatial concepts. Furthermore, substantial variability exists in their ability to recognize and represent geometric structures, such as squares or hexagons, leaving significant room for improvement (Yamada et al., 2024). The construction of cognitive maps—representations of relational structures in tasks or environments—has also been explored. While cognitive maps are essential for human spatial planning and navigation, systematic

evaluations reveal that LLMs often fail in planning tasks, and there is insufficient evidence to support their competence in cognitive map construction (Momennejad et al., 2023).

In summary, while LLMs have made measurable progress in spatial understanding, further advancements are necessary for practical applications in real-world scenarios. Discrepancies and inconsistent findings regarding their spatial representation capacities may stem from the absence of standardized experimental paradigms. To address this, it is essential to compare LLMs' spatial representations with those of humans, using well-established testing paradigms from cognitive science. This approach could provide critical insights into optimizing LLMs' spatial reasoning capabilities while ensuring the scientific rigor and validity of experimental evaluations.

3 Methods

3.1 Spatial representation tasks and datasets generation

The spatial language capabilities of LLMs were examined by requesting the models to describe spatial relationships between given object pairs. Two tasks, adapted from human psychological experiments (Munnich et al., 2001; Hayward and Tarr, 1995), were employed to assess their spatial abilities: (1) generating spatial terms to capture spatial relationships and (2) rating the appropriateness of given statements about object locations in a 2D scene. The procedures for these tasks are as follows.

Spatial Generation Task. In the spatial generation task, LLMs were required to produce spatial terms that described the relationships between two objects on a 2D 7*7 grid (Figure 1). The two objects in each trial were the reference object and the figure object. The reference object was always positioned at the center of the grid, while the figure object could appear in any of the remaining 48 positions, centered in the corresponding cells. Five reference-figure object pairs—"computer-ring", "apple-fish", "bird-tree", "book-pen", and "desk-sofa"—were used to create a diverse dataset. This design resulted in a total of 240 trials (48 positions * 5 object pairs). For each trial, a query prompt was generated using the following template, where [reference], [figure], and [x1, y1] were replaced with specific values for the trial, and [relation] was to be completed by LLMs.